Countercyclical Consumption Taxes

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  - Spending more effective to stimulate output: Multiplier ≈ 1 on impact

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  - Transfers better to boost private consumption, but multiplier $\approx$ MPC on impact

    Kaplan & Violante (2014), McKay & Wolf (2025), Ferriere & Navarro (2025), Ramey (2025), ...

Policy Proposal

- Alternative instrument to mitigate recessions: Consumption tax cuts
  - Proposed in the policy debate, used in some countries: UK 2009, Germany 2020

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    - Novel theoretical equivalence result
  - Crowds in private rather than public consumption

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- HANK model to quantify business cycle stabilization (Work in progress...)
  - Additional distributional gains if recessions affect households unequally (Welfare[2])

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- Roadmap: Theoretical equivalence; Quantitative model

Literature

- Effects of monetary policy and government spending in HANK models

  Kaplan, Moll, and Violante (2018), Hagedorn, Manovskii, and Mitman (2025), Bilbiie (2020), Auclert, Rognlie, and Straub (2023), Ferriere and Navarro (2024), Alves and Violante (2023)

- Optimal fiscal and monetary policy in HANK

  Bhandari, Evans, Golosov, and Sargent (2021), Le Grand and Ragot (2024), McKay and Wolf (2023)
  Broer, Druedahl, Harmenberg, and Oberg (2025), Le Grand, Ragot and Bourany (2025)

- Stabilization and consumption taxes in HANK

  Parodi (2024), Bachmann, Born, Goldfayn-Frank, Kocharkov, Luetticke, & Weber (2025), Bartal & Becard (2025)

- Equivalence results in HANK

  Correia, Nicolini & Teles (2008), Correia, Farhi, Nicolini & Teles (2013), Seidl & Seyrich (2023), Wolf (2024), Wolf (2025)

- Evidence on consumption tax cuts

  Blundell (2009), Benzarti, Carloni, Harju, & Kosonen (2020), Bachmann et al. (2025)

A Theoretical Equivalence

The Simplest Model

- RBC model with no capital, no heterogeneity
- Representative household
  - Values consumption C_t and leisure $1 - N_t$
  - Can save in a risk-free bond A_t
- Representative competitive firm
  - Hires labor in a competitive labor market to produce output Y_t
- Government has to finance public good G_t
  - Uses labor and consumption taxes τ_t^n and τ_t^c
  - Issues public debt $B_t = A_t$ in equilibrium
- Prices: (r_t, w_t) the interest rate and the net-of-taxes wage

Two Expansions

- From steady state with $\tau^c_{ss} = 0$ (wlog), consider two temporary fiscal expansions.

1. Government spending (expansion) stimulus $\left\{\hat{G}_t\right\}_{t=0}^{T}$
   - Fiscal cost $\hat{F}_t \equiv \hat{G}_t - G$
   - Financed with any policy $\hat{\Pi}_t \equiv \left(\hat{\tau}^n_t, \hat{B}_t\right)$ converging back to steady state
   - Output & after-tax prices $\hat{X}_t \equiv \left(\hat{Y}_t, \hat{r}_t, \hat{w}_t\right)$ and consumption $\hat{C}_t$

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2. Consumption tax (cut) stimulus $\left\{\bar{\tau}^c_t < \tau^c_{ss} = 0\right\}_{t=0}^{T}$
   - Fiscal cost $\bar{F}_t \equiv -\bar{\tau}^c_t \bar{C}_t$ with policy $\bar{\Pi}_t \equiv \left(\bar{\tau}^n_t, \bar{B}_t\right)$
   - Output & after-tax prices $\bar{X}_t \equiv \left(\bar{Y}_t, \bar{r}_t, \bar{w}_t\right)$ and consumption $\bar{C}_t$

Equivalence Result

- **Assumption:** Assume log-separable preferences $U(c, n) = \log c - v(n)$.
- **Proposition:** Let the two policies be identical $\bar{\Pi}_t = \bar{\Pi}_t \ \forall t$. Then,
  - The equilibrium $\bar{\tau}_t^c$ is such that $\hat{F}_t = \bar{F}_t \ \forall t$;
  - Output and after-tax prices are identical: $\bar{X}_t = \bar{X}_t \ \forall t$;
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  - The consumption tax cut crowds-in private consumption: $\bar{C}_t = \hat{C}_t + \left(\hat{G}_t - G\right)$
- **Corollary:** Spending and consumption tax shocks generate identical output multipliers

$$M_y^g = M_y^\tau$$

  - The consumption tax cut that replicates the spending stimulus is given by:

$$\bar{\tau}_t^c = -(\hat{G}_t - G)/(\hat{Y}_t - G)$$

Intuition The log case

- A recursive formulation of the household problem reads:

$$V_t(A_t) = \max_{C_t, N_t, A_{t+1}} \{\log(C_t) - v(N_t) + \beta V_{t+1}(A_{t+1})\} \text{ s.t.}$$

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  - Consistent with firms decisions
  - Consistent with government's budget constraint as identical cost
  - Consistent with market clearing if and only if $\bar{C}_t = \hat{C}_t + \left(\hat{G}_t - G\right) = \left(\hat{Y}_t - G\right)$

Extending the Equivalence

- Straightforward extension to aggregate shocks and $\tau^c_{ss} > 0$
  - **Equivalence** of any package $\{\tilde{G}_t, \tilde{\tau}^c_t\}$ delivering the same fiscal cost:
    $\tilde{F}_t \equiv (\tilde{G}_t - G_t) - (\tilde{\tau}^c_t \tilde{C}_t - \tau^c_t C_t)$

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  - Capital in the production function
  - Heterogeneous agents with borrowing constraints
  - Various labor supply arrangements (extensive margin)
  - Nominal rigidities: sticky prices, sticky wages

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- Key assumption: **log preferences**, income and substitution effects cancel out
  - Static margin (labor supply)
  - Dynamic margin (savings decisions)

Details

Beyond log separable preferences

- Simple setup: RA RBC, no capital, one-time shock financed with lump-sum transfers
  1. Flexible prices: r_t so that the Euler equation holds
  2. Nominal rigidities and central bank implementing a peg: constant real rate
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  - (Consumption multipliers: $M_c^g < M_c^\tau \ \forall \sigma$)

Intra vs Intertemporal σ

- Preferences to disentangle σ^n (labor supply) from σ^s (savings)
  - Let $\sigma^n = \partial \log \mathcal{R} / \partial \log c$, with $\mathcal{R} \equiv -u_n(c,n)/u_c(c,n)$
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  - Under Taylor rule: $$M_y^g = \left[\frac{\kappa \Phi_\Pi}{1 + \kappa \Phi_\Pi} \sigma^n + \frac{1}{1 + \kappa \Phi_\Pi} \sigma^s \right] M_y^\tau,$$

    where κ is the slope of the Phillips Curve (small)

Quantitative Model

A HANK Model with Unemployment

- Households with log separable preferences
  - Bond economy with borrowing constraint
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  - Separation and job finding rates function of individual productivity and aggregate state
  - $\Rightarrow$ Unequal incidence across households and increasing in recessions

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  - $\Rightarrow$ Aggregate demand effects
- Government finances spending, transfers, UI benefits with debt, τ^n, τ^c, τ^k

Households **Working households**

- Individual state: asset a, discount factor β, productivity x, and employment $\eta \in \{\ell, u\}$

- Value function when employment "island" $\eta = \ell$

$$V_t(a, x, \ell, \beta; \mu_t) = \max_{c, a'} \left\{ \log c - B \frac{h^{1+\varphi}}{1+\varphi} + \beta \mathbb{E}_t \left[V_{t+1}(a', x', \eta', \beta'; \mu_{t+1}) | x, \beta, \ell \right] \right\} \quad \text{s.t.}$$

$$(1 + \tau_t^c) c + a' = a + y^\ell + y^k - \mathcal{T}_t(y^\ell, y^k) + T_t,$$

$$y^\ell = w_t x h, \quad y^k = r_t a, \quad a' \geq 0, \quad \mu_{t+1} = \Psi(\mu_t).$$

  - AR(1) process for discount factor, productivity and employment status
  - Flat capital and consumption taxes τ^k and τ_t^c, progressive loglinear labor tax (λ_t, τ^ℓ)
    Heathcote, Storesletten, and Violante (2017)

Households **Unemployed households**

- Value function when in unemployment "island" $\eta = u$

$$V_t(a, x, u, \beta; \mu_t) = \max_{c, a'} \{\log c + \beta \mathbb{E}_t \left[V_{t+1}(a', x', \eta', \beta'; \mu_{t+1}) | x, \beta, u\right]\} \quad \text{s.t.}$$

$$(1 + \tau_t^c)c + a' = a + y^k + \mathcal{B}_t(w_t x h) - \mathcal{T}_t(0, y^k) + T_t,$$

$$y^k = r_t a, \quad a' \geq 0.$$

  - Unemployment benefits function of hourly wage

    Kekre (2022)

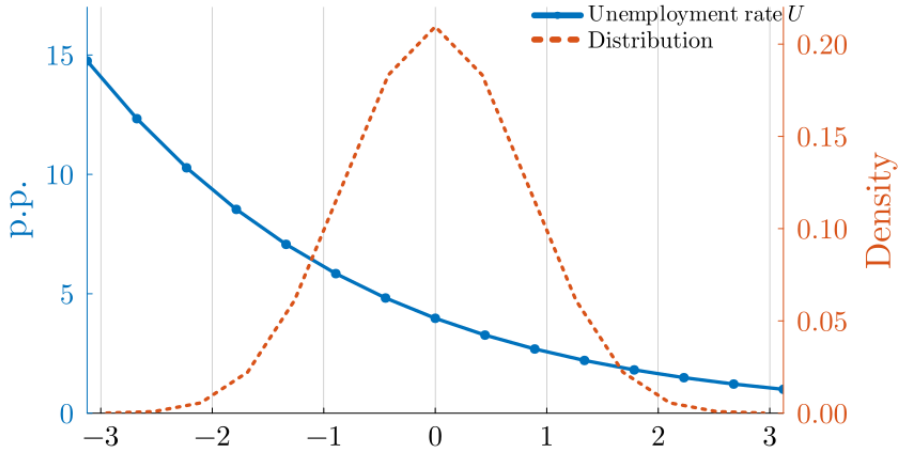
$$\mathcal{B}_t(w_t x h) = \zeta \min\left(\mathcal{R} w_t x h, \overline{ui}\right) + \chi w_t x h$$

    + ζ to match fraction of recipients, $\mathcal{R}$ the replacement rate, $\overline{ui}$ the UI cap
    + χ to capture household labor income received while in unemployment

  - AR(1) process for discount factors, productivity and employment status

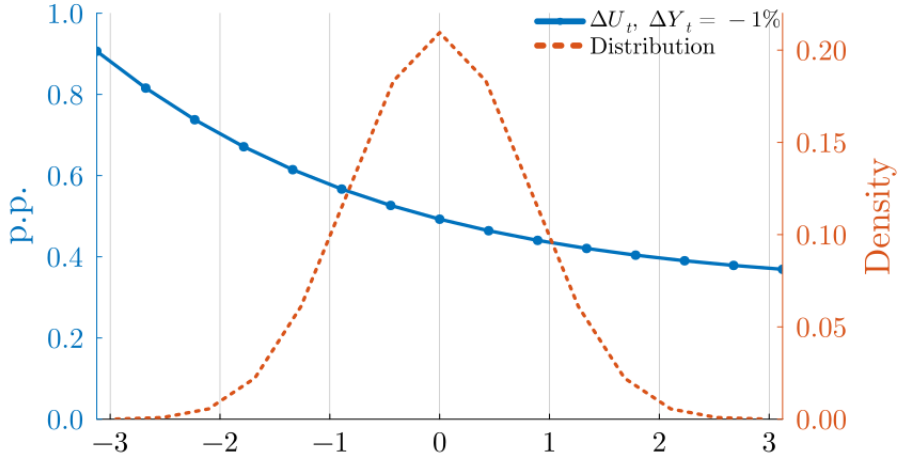
Unemployment Steady State

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Unemployment Business Cycles

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Investigating the Calibration **Household responses**

- Marginal propensities to consume (mpc)

  Parker, Souleles, Johnson, and McClelland (2013), Kaplan and Violante (2014), ...
  Saporta-Eksten (2014), Ganong and Noel (2019)

  – Compute mpc out of a \$500 rebate: average quarterly mpc at 0.20

  – Decline with wealth: from 0.59 to 0.03 from 1st to 5th wealth quintile

Wealth quintile	1	2	3	4	5
Wealth share	0.00	0.01	0.05	0.18	0.76
mpc	0.59	0.19	0.09	0.06	0.03

  – Larger for unemployed at 0.45

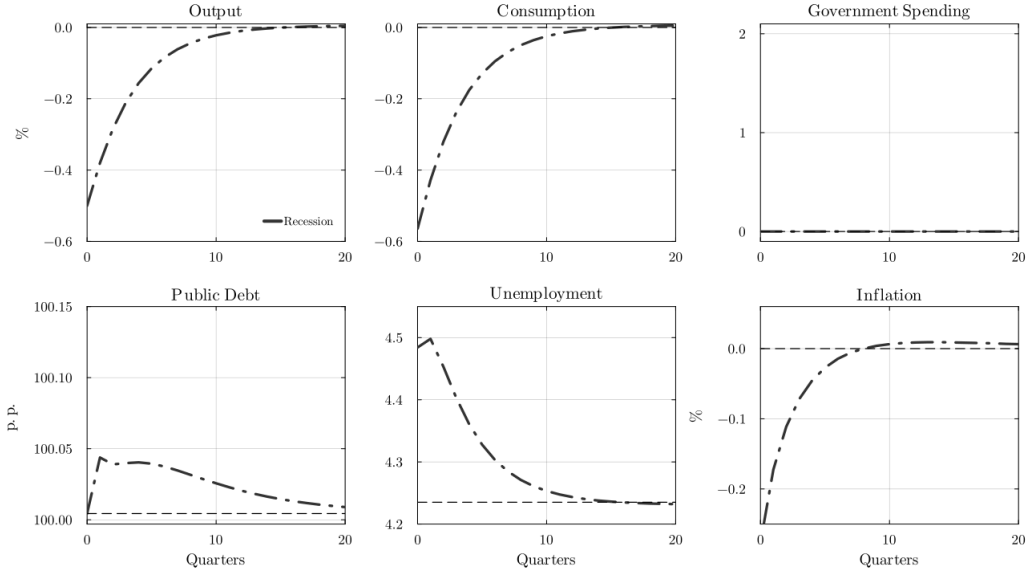
Recession Demand Shock

- Recession induced by a negative demand shock: $(1-\omega_t)u(c_t, n_t)$
  - ω_0 such that $\Delta Y_t = -0.5\%$ on impact
  - Reverts to $\omega = 0$ with persistence $\rho_\omega = 0.75$ at the quarterly level
- Unexpected, transitory, perfect foresight: a 'MIT' shock

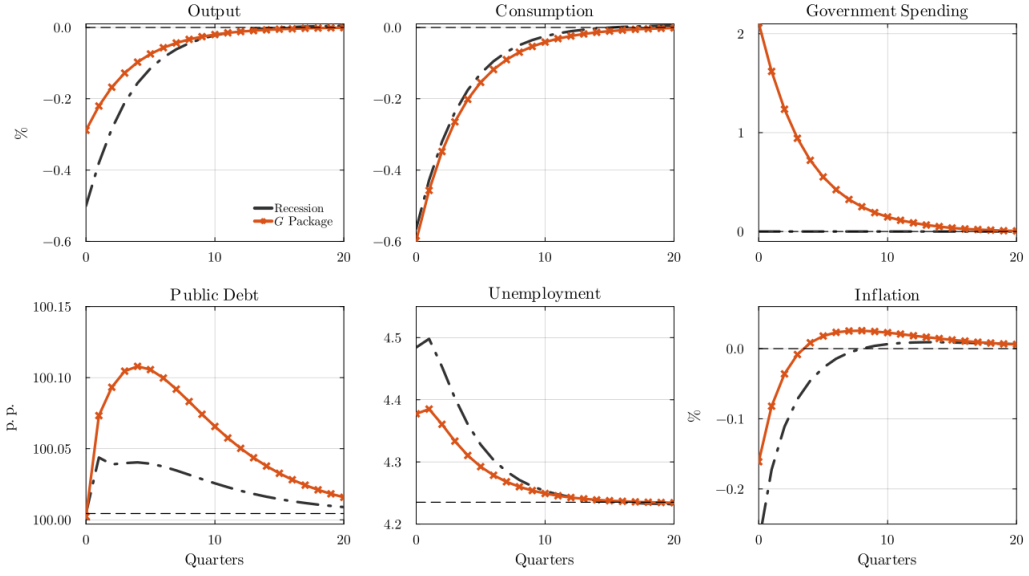
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- Three packages of equal fiscal cost
  - G package: Reduces the output loss by $\approx 50\%$ on impact, persistence ρ_w
  - τ^c package
  - T package

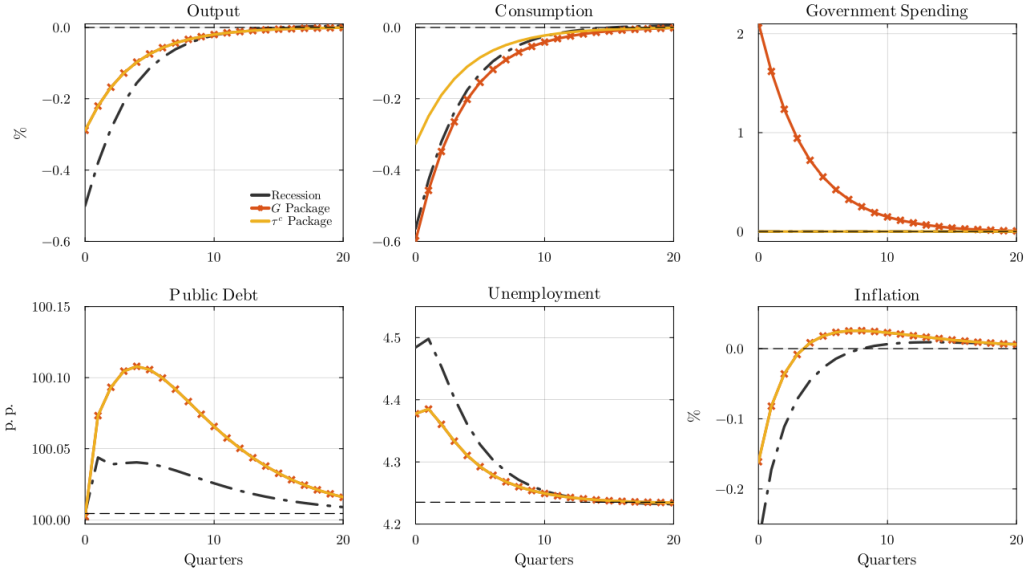
Recession Stabilization



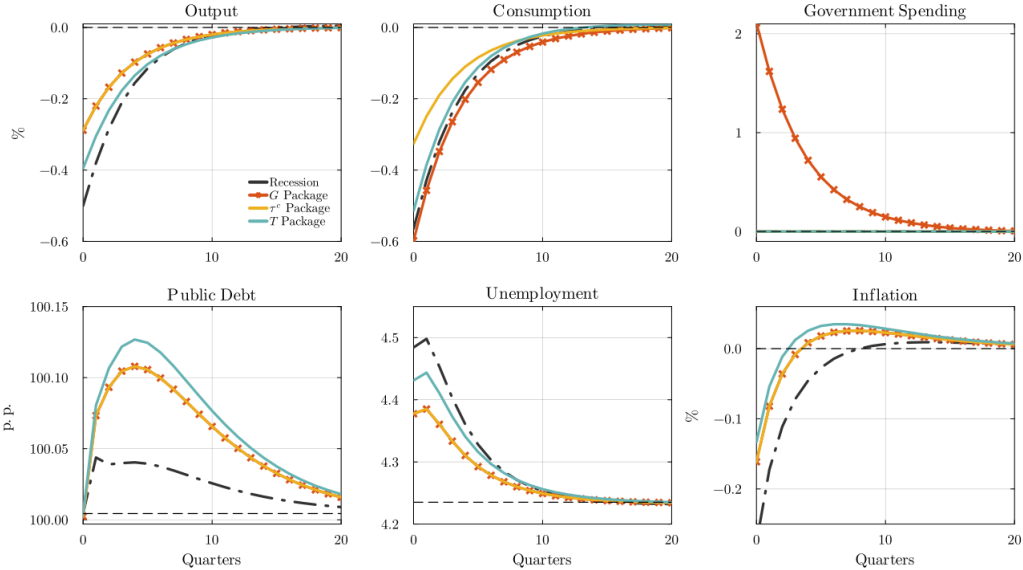
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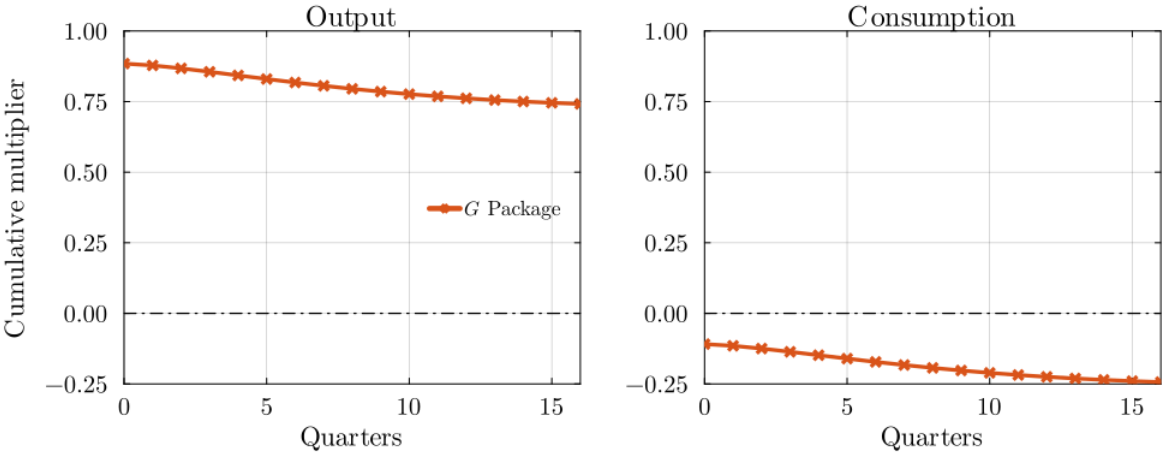


Recession Stabilization

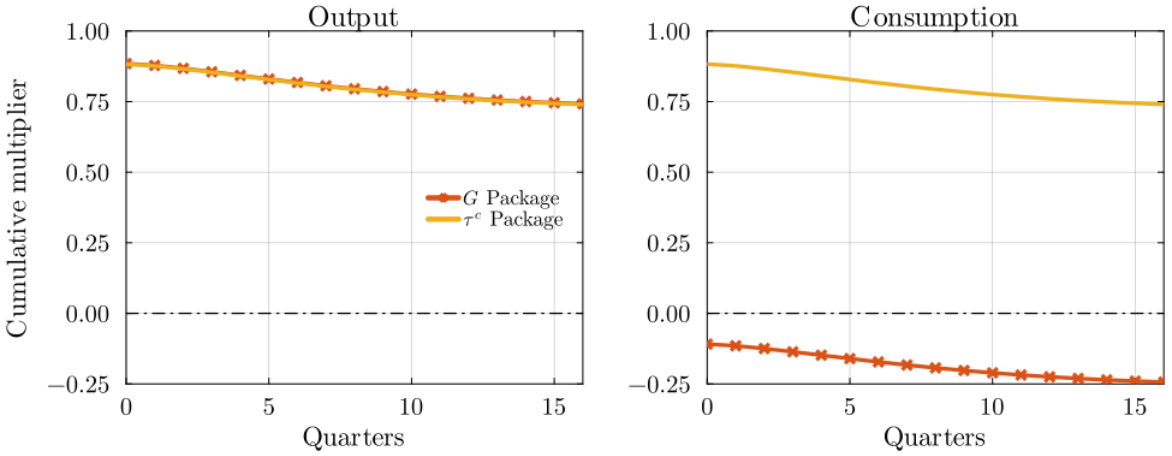


Multipliers

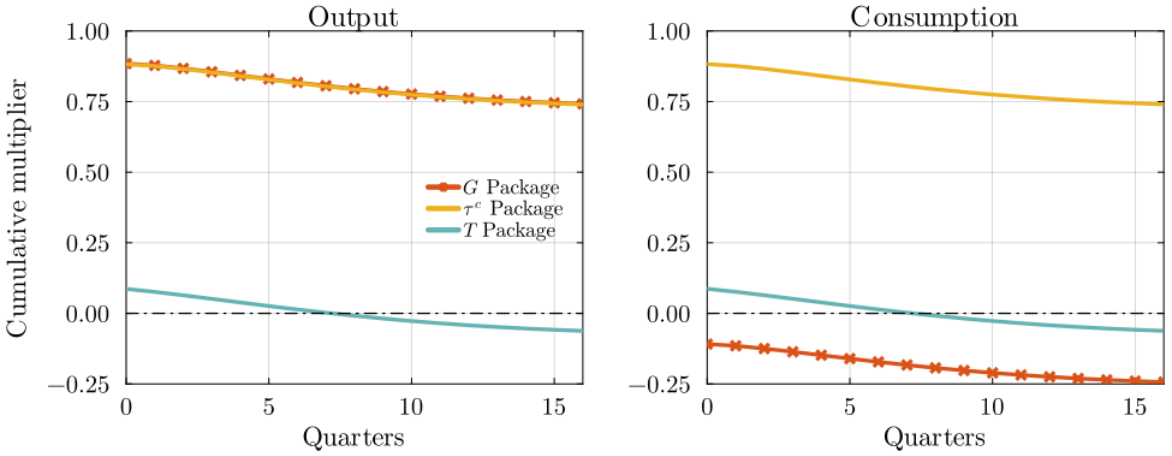
Multipliers



Multipliers



Multipliers



Conclusion

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  - As effective as G to stimulate output, while operating through private consumption only

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- What next?
  - Optimal stabilization: Aggregate shocks and second-order approximations
  - Theoretical comparison to transfers T, quantitative comparison to targeted transfers
  - Valuation of G; baskets of goods

Thank you!

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  - Replicate monetary policy by a combination of $\{\tau_t^c, \tau_t^n, \tau_t^k\}$ in a RANK
  - Seidl & Seyrich (2023): Extension in HANK
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- Wolf (2024)
  - Conditions under which changes in private spending and changes in public spending generate same general equilibrium effects
    - + Same path for aggregate demand
    - + Same path of future taxesr

Return

Extending the Equivalence

- Works with aggregate shocks and non-zero consumption tax
  - Let X_t denote a variable in t in absence of fiscal stimulus;
  - Fiscal costs defined as:

    $\hat{F}_t \equiv (\hat{G}_t - G_t) - \tau_t^c(\hat{C}_t - C_t)$ and $\bar{F}_t \equiv -(\bar{\tau}_t^c - \tau_t^c)\bar{C}_t - \tau_t^c(\bar{C}_t - C_t)$
    - The consumption tax cut that replicates the spending stimulus is given by:

$$\bar{\tau}_t^c = \tau_t^c \frac{\hat{Y}_t - \hat{G}_t}{\hat{Y}_t - G_t} - \frac{\hat{G}_t - G_t}{\hat{Y}_t - G_t}$$

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Extending the Equivalence

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- **Equivalence** of any $\{\tilde{G}_t, \tilde{\tau}_t^c\}$ that deliver the same fiscal cost:

  $\tilde{F}_t \equiv (\tilde{G}_t - G_t) - (\tilde{\tau}_t^c - \tau_t^c)\tilde{C}_t - \tau_t^c(\tilde{C}_t - C_t)$
  - Finance spending shock with consumption taxes $\Rightarrow$ multiplier $= 0$
  - Full crowding-out of private consumption!

Return

Preferences

- Define $\mu \equiv u_c(c, n)$ and $\sigma^s = \frac{\partial \log \mu}{\partial \log c}$
- Define $R \equiv -\frac{u_n(c,n)}{u_c(c,n)}$ and $\sigma^n = \frac{\partial \log R}{\partial \log c}$

Return

Preferences

- King-Plosser-Rebello

$$U(c_t, n_t) = \frac{1}{1-\sigma}\left((c_t(1-n)^{\nu})^{1-\sigma} - 1\right)$$

delivers $\sigma^n = 1$ and $\sigma^s = \sigma$.

- Jaimovich-Rebelo

$$U(c_t, n_t; x_t) = \frac{1}{1-\sigma}\left(c_t - \frac{n_t^{1+\varphi}}{1+\varphi}x_t\right)^{1-\sigma}$$

where $x_t = c_t^{\gamma}$, delivers $\sigma_n = \gamma$ and $\sigma_s = \sigma\frac{\mu(1+\varphi)}{\mu(1+\varphi)-1}$.

- Adjusted CRRA

$$U(c_t, n_t; x_t) = \frac{c_t^{1-\sigma^s}}{1-\sigma^s} - x_t\frac{n_t^{1+\varphi}}{1+\varphi}$$

where $x_t = c_t^{\sigma^n - \sigma^s}$ is a reference point.

Return

Firms and Labor Market

- Standard two-layer structure
  Hagedorn, Manovskii, and Mitman (2025)

Return

Firms and Labor Market

- Standard two-layer structure
  Hagedorn, Manovskii, and Mitman (2025)
- Representative final good producer
- Intermediate good producers indexed by k
  - Linear technology: $y_{kt} = n_{kt}$, flexible prices $\rightarrow$ constant wage w

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Firms and Labor Market

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- Representative final good producer
- Intermediate good producers indexed by k
  - Linear technology: $y_{kt} = n_{kt}$, flexible prices $\rightarrow$ constant wage w
- Unions indexed by j
  - Demand same labor from all working households and aggregate it linearly
  - Nominal rigidities a la Rotemberg

Return

Unions Phillips Curve

- Continuum of unions j demanding labor h_{ji} from households i
  - Labor demand: $n_{jt} = \left(\frac{W_{jt}}{W_t}\right)^{-\epsilon} Y_t$
  - Nominal wage W_{jt} s. to utility cost: $\frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t$
  - Atomistic union: take as given total consumption and total hours of working households

$$J_t(W_{tj-1}) = \max_{W_{jt}} \left\{ \int \left[\frac{W_{jt}}{P_t} \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} - \frac{v'(h_i)}{u'(c_i)} \right] h_j di - \frac{\Theta}{2}\left(\frac{W_{jt}}{W_{jt-1}} - \bar{\Pi}\right)^2 Y_t + \beta J_{t+1}(W_{tj}) \right\}$$

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- Continuum of unions j demanding labor h_{ji} from households i
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- **Phillips curve** collapses to Erceg et al (2000) with no heterogeneity

$$\left(\Pi_t^w - \bar{\Pi}\right)\Pi_t^w + \frac{\epsilon - 1}{\Theta}\frac{w}{\bar{x}_t}\int \frac{(1 - MTR_i)x_i}{1 + \tau_{ct}} di = \frac{\epsilon}{\Theta}\frac{1}{\bar{x}_t}\int \frac{v'(h)}{u'(c_i)} di + \beta\left(\Pi_{t+1}^w - \bar{\Pi}\right)\Pi_{t+1}^w \frac{Y_{t+1}}{Y_t}$$

Return

Government

- Monetary authority follows a Taylor rule: $1 + i_t = (1 + \bar{i}) \left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\phi_\Pi}$
- Fiscal authority faces a standard borrowing constraint

$$G_t + (1 + r_t)D_t + T_t + \int \mathcal{B}_t(w_t x)d\mu_t = D_{t+1} + \int \mathcal{T}_t(y_t^\ell, y_t^k)d\mu_t$$

- Fiscal rule with parameter Φ_D for public debt, λ_t clears the budget constraint

  Uhlig (2010)

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- Fiscal rule with parameter Φ_D for public debt, λ_t clears the budget constraint
  Uhlig (2010)
  - $\Phi_D = 0$ for constant debt, all adjustment in tax level
  - $\Phi_D \to 1$ for constant taxes, all adjustment in debt

    $D_{t+1} = (1 - \phi_D)D + \phi_D \left(\hat{G}_t - \tau^k r_t A_t - \tau_t^c C_t - \mathcal{R}_t^\ell\right)$, where
    - $\hat{G}_t$ denotes total spending, including debt repayments
    - $\mathcal{R}_t^\ell$ denotes fiscal revenues at steady-state labor tax schedule

Return Fiscal rule

Steady State **Households**

- Quarterly model calibrated to liquid wealth

Return

Steady State **Households**

- Quarterly model calibrated to liquid wealth
- Stochastic $\beta \in \{\bar{\beta} - \Delta,\ \bar{\beta},\ \bar{\beta} + \Delta\}$, duration of 50 years

  Krusell and Smith (1998)
  - $\bar{\beta}$ s.t. $r \equiv 3.5\%$ annually
  - Δ s.t. top-quintile liquid wealth $\approx 76\%$ (SCF)

    Ferriere and Navarro (2024)

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  - Δ s.t. top-quintile liquid wealth $\approx 76\%$ (SCF)
    Ferriere and Navarro (2024)
- Labor supply
  - B to match hours $h = 1/3$, Frisch $\varphi^{-1} = 0.4$
- Productivity $(\rho_x, \sigma_x) = (0.989, 0.287)$
  Chang and Kim (2007)

Return

Steady State **Firm and government**

- Technology: $\varepsilon = 7$, $\Theta = 200 \rightsquigarrow$ Phillips curve slope $\varepsilon/\Theta = 0.035$
  Galí and Gertler (1999)

- Government
  - Capital tax $\tau^k = 35\%$, consumption tax $\tau^c = 5\%$
    Labor income tax progressivity $\tau^\ell = 0.06$, transfers to match estimates from CPS
    Chen, Imrohoroglu, and Imrohoroglu (2007), Heathcote, Storesletten, and Violante (2017), Ferriere, Grübener, Navarro, and Vardishvili (2023)
  - Fiscal revenue-over-output 22%, debt $D/Y = 100\%$, spending $G/Y = 11\%$

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  - Fiscal revenue-over-output 22%, debt $D/Y = 100\%$, spending $G/Y = 11\%$
  - Unemployment benefits: $\zeta = 0.4$, $\mathcal{R} = 0.5$, $\overline{ui} = 60\%\bar{y}$
    $\chi = 0.24$ to match $C_u/C_e \approx 70\%$
    Kekre (2022), Gorn and Trigari (2024)

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    Kekre (2022), Gorn and Trigari (2024)
  - Automatic responses of inflation and debt: $\Phi_\Pi = 1.5$, $\Phi_D = 0.75$

Return

Unemployment Steady State and Business Cycles

- Job finding rates and separation rates across hourly wage distribution

  Mueller (2017)

Unemployment Steady State and Business Cycles

- Job finding rates and separation rates across hourly wage distribution
  Mueller (2017)

- Steady State
  - Job finding rates constant, separation rates falling in hourly wage/productivity x
  - Average unemployment rate at 4.3%

Unemployment Steady State and Business Cycles

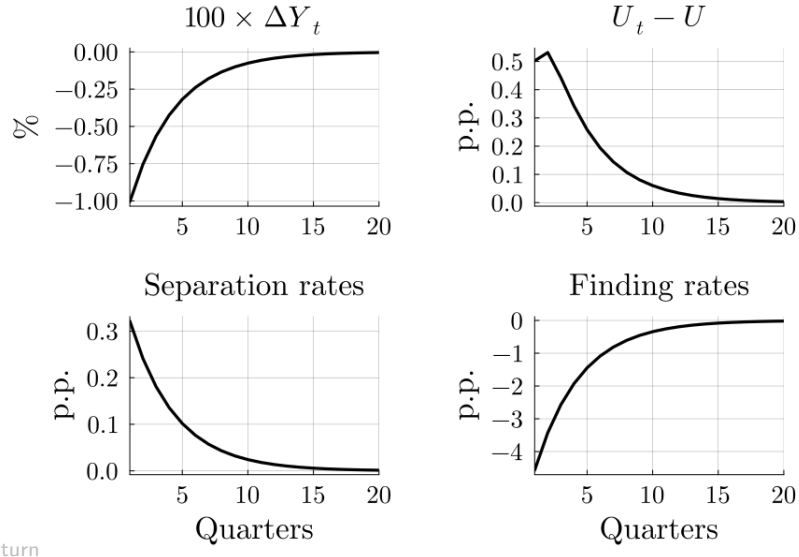
- Job finding rates and separation rates across hourly wage distribution
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- Steady State
  - Job finding rates constant, separation rates falling in hourly wage/productivity x
  - Average unemployment rate at 4.3%

- Okun's law: Okun coefficient $c_{OK} = 0.5$
  Ball, Leigh, and Loungani (2017)
  - Job finding rates decrease (a lot) equally across households
  - Job separation rates decrease (a bit), higher elasticity for high-x households
    + Functional forms: additive fall in separation rates in recession delivers the pattern

Unemployment

Business Cycle: Okun's law



Return

Fiscal Rule

- Public debt adjusts as a function of Φ_D

$$D_{t+1} = (1-\phi_D)D + \phi_D \left(\hat{G}_t - \tau^k r_t A_t - \tau_t^c C_t - \mathcal{R}_t^\ell\right), \text{ where}$$

  - $\hat{G}_t$ captures total government expenditures, including debt repayments

    $\hat{G}_t = G_t + T_t + \mathcal{U}_t + (1+r_t)D_t$

  - $\mathcal{R}_t^\ell$ captures fiscal revenues at steady-state labor tax schedule

    $\mathcal{R}_t^\ell = w_t L_t - \lambda \int (w_t x h_t(a,x,\eta,\beta))^{1-\tau^\ell} d\mu_t(a,x,\eta,\beta)$

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Steady State Households

- Quarterly model calibrated to liquid wealth

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- Stochastic $\beta \in \left\{\bar{\beta} - \Delta,\ \bar{\beta},\ \bar{\beta} + \Delta\right\}$, duration of 50 years
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Steady State Households

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- Stochastic $\beta \in \{\bar{\beta} - \Delta,\ \bar{\beta},\ \bar{\beta} + \Delta\}$, duration of 50 years
  Krusell and Smith (1998)
  - $\bar{\beta}$ s.t. $r \equiv 3.5\%$ annually
  - Δ s.t. top-quintile liquid wealth $\approx 90\%$ (SCF)
- Labor supply decisions
  - B to match employment $\approx 78\%$
    Jang, Sunakawa, and Yum (2023)
  - ρ_h to match average annual labor elasticity of ≈ 0.3
    Ferriere and Navarro (2024)

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Return

Steady State **Firm and government**

- Technology: $\varepsilon = 7$, $\Theta = 200 \rightsquigarrow$ Phillips curve slope $\varepsilon/\Theta = 0.035$

  Galí and Gertler (1999)

- Dividends redistributed linearly in x: $d_t(x) = \bar{d}_t x$

  Farhi and Werning (2019)

Return

Steady State Firm and government

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- Dividends redistributed linearly in x: $d_t(x) = \bar{d}_t x$
  Farhi and Werning (2019)

- Government
  - Capital tax $\tau_k = 35\%$, labor income tax progressivity $\tau_\ell = 0.1$
    Chen, Imrohoroglu, and Imrohoroglu (2007), Heathcote, Storesletten, and Violante (2017), Ferriere, Grübener, Navarro, and Vardishvili (2023)
  - Spending $G/Y = 10\%$, transfers $T/Y = 8\%$, debt $D/Y = 100\%$

Return

Steady State Firm and government

- Technology: $\varepsilon = 7$, $\Theta = 200 \rightsquigarrow$ Phillips curve slope $\varepsilon/\Theta = 0.035$
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  - Unemployment benefits: $\zeta = 0.4$, $\mathcal{R} = 0.5$, $\overline{ui} = 60\%\bar{y}$
    $\chi = 0.15$ to match $C_u/C_e \approx 75\%$
    Kekre (2022), Gorn and Trigari (2024)

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Steady State **Firm and government**

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    Kekre (2022), Gorn and Trigari (2024)
  - Automatic responses of inflation and debt: $\Phi_\Pi = 1.5$, $\Phi_D = 0.75$

Return

Steady State Unemployment

- Job finding rates and separation rates across hourly wage distribution

Return

Steady State **Unemployment**

- Job finding rates and separation rates across hourly wage distribution

- Job finding rates are constant in the distribution
  Mueller (2017)
  - Monthly finding rate of $0.32 \Rightarrow \pi_\eta(\ell|u) = 0.691$

Return

Steady State **Unemployment**

- Job finding rates and separation rates across hourly wage distribution

- Job finding rates are constant in the distribution

  Mueller (2017)

  – Monthly finding rate of $0.32 \Rightarrow \pi_\eta(\ell|u) = 0.691$

- Separation rates are falling in hourly wage/productivity x

  Mueller (2017)

  – Monthly separation rates of $\approx 1.4\%$ and 0.7% below and above median, respectively

  $\Rightarrow$ $\pi_\eta(u|\ell, x) = \phi_0 x^{\phi_1}$, with $\phi_0 = 0.029$ and $\phi_1 = -0.446$

Return

Steady State **Unemployment**

- Job finding rates and separation rates across hourly wage distribution

- Job finding rates are constant in the distribution

  Mueller (2017)

  – Monthly finding rate of $0.32 \Rightarrow \pi_\eta(\ell|u) = 0.691$

- Separation rates are falling in hourly wage/productivity x

  Mueller (2017)

  – Monthly separation rates of $\approx 1.4\%$ and 0.7% below and above median, respectively

  $\Rightarrow$ $\pi_\eta(u|\ell, x) = \phi_0 x^{\phi_1}$, with $\phi_0 = 0.029$ and $\phi_1 = -0.446$

- Average unemployment rate at 4.3% with unequal incidence in the distribution

Return

Unemployment and the Business Cycle

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  - Okun coefficient $c^{OK} = 0.5$

    Ball, Leigh, and Loungani (2017)

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- Job separation rates decrease with ΔY_t
  - *Elasticity* of separation rates to aggregate unemployment larger for above-median workers
    Mueller (2017)
    + Homogeneous additive increase in separation rates

Unemployment and the Business Cycle **Okun's law**

- Finding and separation rates distribution depend on U_t Mueller (2017)
  - Finding rate responds homogeneously (level, independent of x) to ΔY_t

$$\pi_{\eta,t}(\ell \,|\, u) = \pi_{\eta}(\ell \,|\, u) + \bar{\phi}_e \, \Delta Y_t$$

  - Separation rate elasticity increases with ΔY_t

$$\pi_{\eta,t}(u|\ell, x) = \pi_{\eta}(u|\ell, x) - \bar{\phi}_u \Delta Y_t x^{-\phi_{u,x}}$$

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  - Joint calibration:
    - $\bar{\phi}_e$ s.t. finding elasticity to $U \approx -0.6$
    - $\phi_{u,x} = 0$ elasticity of separation rates larger for above-median workers
    - $\bar{\phi}_u = 0.32$ to get $c_{OK} = 0.5$

Return

Transfers vs. Consumption Taxes

- Wolf (24): $dC_G^{GE} = dC_T^{GE}$ under GHH Preferences, $dC_G^{GE} \approx dC_T^{GE}$ away from GHH [1]
- This paper: $dY_G \approx \sigma dY_\tau$, thus $dC_\tau = \sigma^{-1}(dC_G + 1)$ [2]
- Putting everything together:

$$dC_T = dC_T^{PE} + dC_T^{GE} = dC_T^{PE} + dC_G^{GE} = dC_T^{PE} + dC_G \text{ using [1]}$$

- Rearranging, we get that $dC_T > dC_\tau$ only if

$$dC_T^{PE} + dC_G > dC_\tau = \sigma^{-1}(dC_G + 1) \text{ using [2]}$$

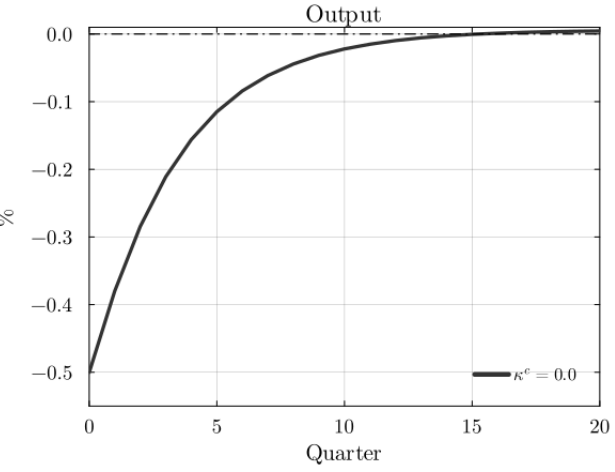
$\Rightarrow dC_T > dC_\tau \Leftrightarrow dC_T^{PE}[= \text{MPC}^{\text{AGG}}] > \frac{1}{\sigma} + \frac{1-\sigma}{\sigma} dC_G^{GE}$

Return

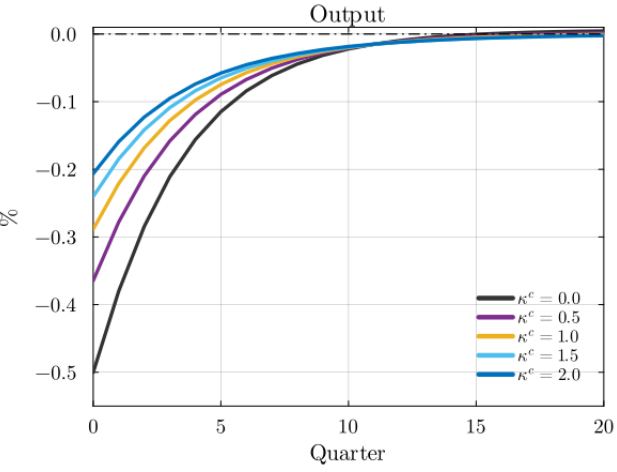
Recession Automatic Stabilizer

- Shall we use consumption taxes as an automatic stabilizer?
- Define a systematic rule: $\tau_t^c = \tau_c + \kappa^c \Delta Y_t$
  - κ^c captures the response of τ^c to the cycle
- Vary $\kappa^c \in \{0, 0.5, 1, 1.5, 2\}$

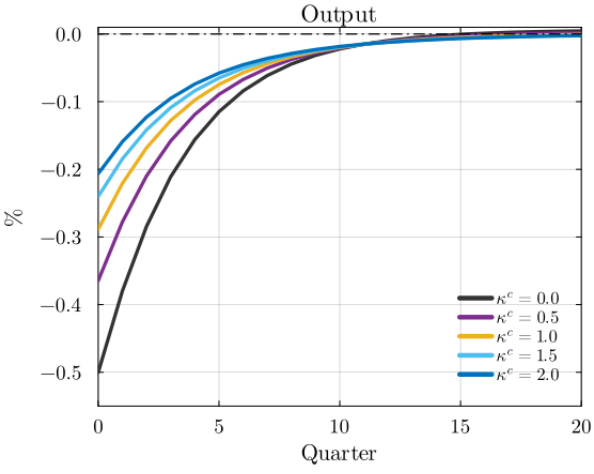
Recession Automatic Stabilizer

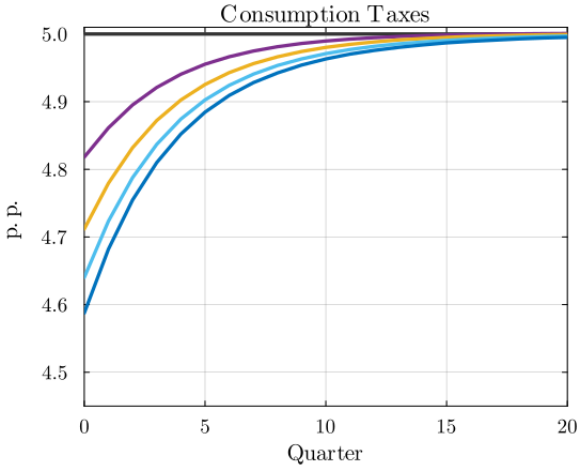


Recession Automatic Stabilizer



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